

Vedant Saxena

Computer Science Professional — AI/ML & IoT Specialist

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Professional Summary

Innovative Computer Science professional with deep expertise in AI/ML, IoT systems, Digital Twins, and full-stack development. Proven research track record with **12 filed patents** (2 under grant review) and multiple international publications in intelligent healthcare and smart infrastructure. Experienced in architecting production-ready Generative AI solutions and scalable cloud applications. Combines rigorous academic research with hands-on engineering capabilities.

Education

- Pursuing **B.Tech in Computer Science Engineering**, *Graphic Era Hill University*, Dehradun, CGPA: 8.21
- 2023 **Diploma in Computer Science Engineering**, *Graphic Era Hill University*, Dehradun, CGPA: 8.78
- 2020 **High School**, *LPS*, Lucknow, 83.5%

Experience

- Oct 2024 – **Engineering Intern**, *DataCulture Technologies*, Remote/Hybrid
- Feb 2025
- Architected and deployed scalable Generative AI workflows and automated ad-bot pipelines, significantly reducing manual operational capabilities.
 - Engineered full-stack web systems, optimizing both front-end and back-end modules to enhance system performance and responsiveness.
 - Collaborated within cross-functional agile teams to translate complex AI research into production-ready solutions.

Patents (12 Filed, 2 Under Review)

- Grant Review **IoT-Enabled Canal Flow Monitoring and Blockage Detection**
- Grant Review **Patient-Specific Digital Twin for Therapeutic Forecasting**
- Filed AI-Powered Voice-Assisted Elevator Call System for Smart Cities
- Filed Smart Charging Surface with Temperature-Sensitive Cut-off
- Filed Privacy-Preserving EHR Anonymization Using MCMC + SHA-256
- Filed Smart AI-Powered Spectacles for Emotional Regulation
- Filed Adaptive Digital Twin for Tailings Dam Safety
- Filed Ethanol-Corrosion Detection Sensor System (E5/E10)
- Filed Compact Mesh + P2P Emergency Communication Device
- Filed Heat Detection and Alert System for Lighting Devices
- Filed Wearable Throat Vibration Device for OM Chant Detection
- Filed Localized Blood-Based Drug Reaction Analysis System

Publications

Conference Papers (Accepted)

- Accepted **Assessment of Channel Estimation Algorithms for mmWave MIMO Systems in 6G Networks**
- Accepted **UAV Integration with mmWave 6G Networks for Coverage and Capacity Enhancement**
- Accepted **Federated Learning-Based Privacy-Preserving QoS Optimization in 6G Networks**
- Accepted **Neuromorphic Vision Systems: Bio-Inspired Image Processing**
- Accepted **Evaluating Quantum vs Classical ML under Classical Simulation**

Book Chapters (Under Review)

- Review **IoT-Enabled EEG-Based Cognitive Load Analyzer**
- Review **Gender Classification Using Sternum Anatomy with Neuromorphic Computing**

Key Projects

- AI/IoT **AI-Powered Voice-Assisted Elevator Call System**
Developed a touchless elevator system integrating Alexa, Google Assistant, and Siri using a hybrid BLE + cloud-edge architecture, enhancing accessibility for smart cities.
- Full Stack **NoirBoard Productivity Suite**
Built a privacy-first productivity tool (Chrome extension + Web App) using React and Tailwind CSS. Features include real-time autosave sync for tasks, goals, and expense tracking.
- Algorithms **Smart Traffic Light System**
Designed a real-time traffic prioritization system for emergency vehicles using Dijkstra's algorithm. Implemented a logic engine in C++ with a visualization dashboard in Next.js.
- Systems **FileZap - Cross-Platform File Sharing**
Created a high-speed, LAN-based file transfer system using Node.js and WebSockets. Features a multi-platform CLI and a QR-based browser interface for instant sharing.

Additional ML & Data Science Projects

- Healthcare **Breast Cancer Classification**: Scikit-learn based tumor classification.
- Healthcare **Breast Cancer Detection**: CNN using TensorFlow/Keras.
- Healthcare **COVID-19 X-Ray Detection**: PyTorch (ResNet/DenseNet + Grad-CAM).
- Analytics **Diabetes Prediction**: PySpark MLlib distributed pipeline.
- Healthcare **Coronary Artery Disease Diagnosis**: Feature-engineered ML system.
- Bio-Informatics **DNA Classification**: K-mer + LSTM-based sequence model.
- Healthcare **Heart Disease Classification**: SVM/Random Forest optimized model.
- Analytics **Heart Failure EDA**: Visual exploration of clinical datasets.
- Statistics **Hypothesis Testing**: Drug safety analysis using T-test and chi-square.
- Deep Learning **Diabetes Prediction Using MLP**: ANN model using Keras.
- Analytics **Happiness vs Country Analysis**: Correlation + regression insights.

Technical Skills

- Languages C++, Python, Java, JavaScript, TypeScript, R
- Frameworks Node.js, React, Next.js, Flask, Tailwind CSS, Spring Boot
- Databases MySQL, MongoDB
- Core Concepts Data Structures & Algorithms, AI/ML, Generative AI, System Design, Security Analysis

Tools & OS Git, Unix/Linux, AWS (Architecture), PySpark

Certifications & Achievements

SkillEcted React JS, Spring Boot API, Python Coffee System, JavaScript Game Development

Forage AWS Architecture, Software Engineering Simulation, Red Bull Sales Simulation

Coursera Fake News ML, R Programming, Machine Learning Projects

Udemy PHP + MySQL (5 Projects)

GFG/Apna 160 Days Program; Apna College Java DSA

Events Chandrayaan-3 Quiz, SRCC Trading Challenge, GEHU Esports

Interests

Artificial Intelligence, ML Research, Neuroscience-Inspired Computing, Cloud Systems, Problem Solving.